

Title

Short-Term Traffic Speed Prediction Using PeMS Data

1. Problem Statement

Traffic congestion is a major issue in urban freeway systems, leading to increased travel time, fuel consumption, and environmental impact. Accurate short-term traffic prediction can help improve traffic management and decision-making.

The goal of this project is to develop a machine learning model that predicts freeway traffic speed 30 minutes into the future using historical traffic sensor data.

2. Data Description

This project uses data from the California Performance Measurement System (PeMS), which provides real-time and historical traffic data collected from roadway sensors.

The dataset includes:

- Timestamp (5-minute intervals)
- Station ID (traffic sensor location)
- Traffic flow (vehicles per unit time)
- Occupancy (percentage of time sensor is occupied)
- Speed (mph)

Additional derived features include:

- Hour of day
- Day of week

The dataset used for modeling contains over 20,000 samples, with a larger raw dataset (~400MB) processed during data collection.

3. Problem Type

This is a **supervised regression problem**, where the target variable is:

- **Speed 30 minutes ahead (speed_30min_ahead)**

The goal is to minimize prediction error using metrics such as:

- Mean Absolute Error (MAE)
 - Root Mean Squared Error (RMSE)
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4. Approach

The project will follow a standard machine learning pipeline:

Data Processing

- Clean raw PeMS data
- Handle missing values
- Create time-based features
- Generate prediction targets using time shifting

Modeling

- Start with baseline model (Linear Regression)
- Compare with advanced models:
 - Random Forest
 - Gradient Boosting (XGBoost)

Evaluation

- Use time-based train/test split
 - Evaluate using MAE and RMSE
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5. Initial Findings

Preliminary modeling shows that traffic speed has strong temporal correlation, meaning current speed is highly predictive of near-future speed.

The baseline model achieved low error due to this correlation, indicating that traffic conditions change gradually over time.

6. Future Work

To improve model performance and realism, the following enhancements will be explored:

- Predict longer time horizons (e.g., 60 minutes ahead)
- Incorporate additional datasets (e.g., weather or incidents)
- Experiment with advanced models
- Build a deployment pipeline for real-time predictions

7. Project Value

This project demonstrates the ability to:

- Work with large real-world datasets
- Perform data cleaning and feature engineering
- Build and evaluate machine learning models
- Design a scalable prediction pipeline